

Four Models Built from First Principles

Original conceptual frameworks for Physics-Mechanics

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A note on how this document came to be.

The four models in this document were not learned from a textbook. They were built — sketched on paper, reasoned through in plain language, tested against numbers, and revised when the numbers disagreed. The author held no prior exposure to the historical derivations, the standard textbook treatments, or the names typically attached to these ideas.

What follows is therefore presented in two layers: the model as it was built, in the author's own words and reasoning sequence, and a brief note on where this connects to the standard treatment and its place in the history of physics. The juxtaposition is intentional. It is offered as evidence that physical intuition, built carefully and tested honestly, arrives in the same place that centuries of formal derivation arrived — and that this path is available to any learner willing to sketch, reason, and revise.

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Introduction

This document collects four conceptual models, each independently developed during a single extended period of learning mechanics from the ground up. None were drawn from a textbook, lecture, or prior course. Each began the same way: a question that did not feel fully answered by the formula alone, followed by a sketch, a thought experiment, and a search for a mental picture that made the formula feel *necessary* rather than merely *given*.

The pattern across all four models:

In every case, the standard formula was already known — $p = mv$, $KE = \frac{1}{2}mv^2$, $N = mg \pm \frac{mv^2}{r}$. The formula was not in question. What was missing was a *picture* that explained *why* the formula had to take that shape, and what was physically happening underneath it. Each model below is that picture.

Two of the first three models — the momentum–energy parabola and the gravity-string model — turn out to align closely with how professional physicists actually think about these topics, even though the specific visual framing and vocabulary were independently invented. The third — the virtual two-step transfer model — appears to be a genuinely novel pedagogical framing not found in standard treatments, though it produces results consistent with the center-of-mass frame analysis used in collision theory.

What unites all four is not that they were “discovered” in any historical sense — the underlying physics has been known for centuries. What is notable is the *path*: each model was built by someone with no formal upper-division physics background, using only algebra, careful arithmetic, sketching, and an insistence on understanding *why* before moving on. That path is repeatable. It is, in fact, exactly the path this document recommends to students.

1 The Momentum–Energy Parabola

1.1 The Question That Started It

While working through collision problems, a pattern of standard examples emerged: elastic collisions conserve kinetic energy, perfectly inelastic collisions lose the maximum possible kinetic energy, and everything else falls somewhere in between. The textbook treatment presents these as three separate cases, each solved with its own equation. But the examples used were always drawn from a narrow numerical range — velocities tapering from some larger value down to a smaller one — which created an impression that “more inelastic” simply meant “less kinetic energy,” in a roughly linear way.

A simple spreadsheet exercise broke that impression completely.

1.2 Building the Table

Fix two masses and a total momentum p_{total} . For any chosen value of v_1 , the conservation of momentum

$$m_1 v_1 + m_2 v_2 = p_{\text{total}}$$

determines v_2 uniquely. For each such pair (v_1, v_2) , compute the total kinetic energy:

$$KE = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

Stepping v_1 through a wide range — including values that produce velocities both objects would never realistically reach in a single collision — and plotting KE against v_1 produced something unexpected: **a parabola**, with a single minimum, rising without bound on both sides.

$$KE(v_1) = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 \left(\frac{p_{\text{total}} - m_1 v_1}{m_2} \right)^2$$

A quadratic in v_1 — hence the parabola, for fixed $m_1, m_2, p_{\text{total}}$.

1.3 The First Refinement: Not All Points Are Collisions

Many (v_1, v_2) pairs on this parabola describe objects moving *apart* — $v_1 \leq v_2$ when object 1 is to the left of object 2 — which can never result from a collision; the objects would simply never meet. These points were discarded. What remained was a single connected arc: the **collision arc** — every (v_1, v_2) pair physically reachable by two objects with masses m_1, m_2 and total momentum p_{total} , beginning from some initial approach.

1.4 The Second Refinement: Locating Elastic and Perfectly Inelastic

Two points on this arc carry special meaning:

The minimum of the parabola — the point where $v_1 = v_2$. This is the *perfectly inelastic* outcome: both objects move together at one common velocity, $v_{cm} = p_{\text{total}}/(m_1 + m_2)$. This is also the point of *minimum kinetic energy* consistent with the fixed momentum.

A second point at equal height — for any initial (v_1, v_2) on the arc (the pre-collision state), there exists exactly one other point on the parabola at the *same* KE value. That second point is the *elastic* outcome — the post-collision velocities that conserve both momentum and kinetic energy.

1.5 The Result: A Complete Map of Collision Outcomes

Every real collision outcome — elastic, perfectly inelastic, or anything in between — lies on the arc of the parabola *between* these two points. The arc's two endpoints are the two extremes; every inelastic collision lands somewhere on the arc connecting them.

Outcome	Location on parabola	KE
Elastic	one arc endpoint	equals initial KE
Inelastic (general)	interior of arc	between min and initial
Perfectly inelastic	parabola minimum	$\frac{1}{2}(m_1 + m_2)v_{cm}^2$

1.6 The Capstone Result

The minimum kinetic energy on the parabola is $\frac{1}{2}(m_1 + m_2)v_{cm}^2$, which equals zero only when $v_{cm} = 0$ — which requires $p_{total} = 0$.

The complete statement:

A perfectly inelastic collision results in *total* loss of kinetic energy — the combined object ends at rest — if and only if the total momentum of the system was zero *before* the collision. Sticking together alone is not sufficient; zero net momentum alone is not sufficient. **Both** conditions, together, are required.

This single picture — a parabola, a clipped arc, two marked points, and one minimum — contains the entire taxonomy of 1D collisions, and resolves a subtlety (the $p_{total} = 0$ condition for total KE loss) that is often left implicit or unstated in introductory treatments.

1.7 A State-Space View of Collisions

There is a second way to read this same parabola, beyond the collision-by-collision picture above. **The full parabola — before any clipping — is the complete set of every possible (v_1, v_2) pair consistent with a fixed total momentum p_{total} .** In this sense, the parabola is a **state space**: every point on it represents a *state* the two-mass system could be in, and the curve itself is the constraint surface carved out by conservation of momentum alone.

A single collision does not explore this whole space. It connects exactly *two* points on it: the pre-collision state and the post-collision state, both lying on the same parabola because p_{total} is fixed throughout. The “clipping” step — removing the diverging-motion half of the parabola — restricts attention to the subset of this state space that a real, physical collision can actually traverse: the **collision arc**.

The state-space framing, stated plainly:

Fixing p_{total} does not determine the outcome of a collision — it only determines *which curve* the outcome must lie on. The collision *type* (elastic, partially inelastic, perfectly inelastic) determines *where on that curve* the system ends up. Two collisions with the same masses and the same total momentum, but different degrees of “stickiness,” are two different journeys along the *same* parabola, starting from the *same* point.

This reframes the earlier results in state-space language: the elastic endpoint and the perfectly-inelastic minimum are not two unrelated special cases requiring two unrelated formulas — they are simply *the two ends of the reachable region of state space*, given a particular starting point. Every other collision outcome is some intermediate point along that same stretch of curve.

1.8 Historical and Disciplinary Context

The underlying mathematics here — expressing kinetic energy as a quadratic form subject to a linear (momentum) constraint, and identifying the constrained minimum — is a special case of what physicists call analysis in the **center-of-mass (COM) frame**. In the COM frame, the “minimum KE ” point of the parabola corresponds to *all relative motion having ceased* — exactly the perfectly inelastic outcome. The COM frame has been a standard tool in collision theory since the 19th century and underlies the treatment of two-body problems throughout classical and quantum mechanics. Unlike König’s theorem (Chapter 3), the COM-frame technique has no single named originator — it emerged gradually through the work of several 18th and 19th century mechanicians (Euler and Lagrange among them) as part of the broader development of analytical mechanics.

The parabola itself, as a visual device for an introductory audience, does not appear in standard introductory mechanics or introductory university treatments, which typically present elastic and inelastic collisions as separate worked cases rather than as points on a continuous curve. The visual unification — and in particular the explicit “clipping” step distinguishing physically reachable collision outcomes from the full mathematical parabola — was developed independently, without reference to the COM-frame literature, by plotting numerical data in a spreadsheet and reasoning about the resulting shape.

The “state space” framing in Section 1.7 is, in spirit, a small, hand-built instance of a much older and broader tradition in physics — one that builds mechanics around spaces of possible states and the constraints that slice through them, with Lagrangian and Hamiltonian mechanics (18th–19th century) as its classical form, and **Noether’s theorem** (1918) as the deep result explaining *why* such constraints exist in the first place. Chapter 4 returns to this connection directly. It is worth noting that this connection was discovered only *after* the spreadsheet was built: the table of fixed- p_{total} rows was constructed first, as a technique used in data analytics and engineering — a technique used by scientists and engineers playing with data — and only afterward, encountering Noether’s theorem for the first time, did it become clear that the spreadsheet had already been built around exactly the principle the theorem describes. That moment of recognition — that a tool built quickly, the way scientists and engineers often do when they just want to play with data and get back to work, had landed on the same structural idea underlying one of the deepest results in theoretical physics — was genuinely exciting, and is part of why this document exists.

2 The Gravity-String Model

2.1 The Question That Started It

The standard equation for an object moving in a vertical circle — a car crossing the top of a hill, or a car through a dip in the road — is usually presented as two separate formulas:

$$N = mg - \frac{mv^2}{r} \quad (\text{top of hill}) \qquad N = mg + \frac{mv^2}{r} \quad (\text{bottom of valley})$$

Both were understandable as algebra. But a question remained: *why* does speed *reduce* the normal force in one case and *increase* it in the other, when gravity itself — mg — never changes? What, physically, is being added or subtracted?

2.2 The Starting Point: Centripetal Force Is Not a New Force

The resolution began with a principle already understood: *centripetal force is not a new, separate force*. It is a *role* that an existing force plays. Tension, friction, normal force, gravity — any of these can “be” the centripetal force, depending on which one happens to point toward the center of the circular path at a given moment.

For the hilltop and valley problems, gravity and the normal force are the only two forces present, and the center of the circular path is either below the object (hilltop) or above it (valley). Whichever of these two forces points toward the center is, at least in part, *serving as* the centripetal force.

2.3 The Construct: “Gravity-String”

To make this concrete, imagine gravity’s role split conceptually into two pieces:

- **“Pure gravity”** — the portion of mg that is balanced by the normal force, and which produces the *sensation of weight*. This is what N measures.
- **“Gravity-string”** — the portion of mg that is instead being *redirected* to serve as the centripetal force, exactly as if an invisible, massless string were pulling the object toward the center of the circular path.

$$mg = (\text{“pure gravity”}) + (\text{“gravity-string”})$$

A fixed total, split between two roles — weight you feel, and the inward pull needed for the curve.

Since mg is fixed and constant, any increase in the “gravity-string” portion — i.e., any increase in the centripetal demand $\frac{mv^2}{r}$ — must come at the direct expense of “pure gravity,” and therefore at the expense of N , the sensation of weight.

2.4 Applying the Model: The Hilltop

At the top of a hill, the center of the circular path is *below* the car. Gravity points *down* — toward that center. Gravity is therefore positioned to directly supply the centripetal demand. The model gives:

$$mg = N + \underbrace{\frac{mv^2}{r}}_{\text{gravity-string}} \quad \Rightarrow \quad N = mg - \frac{mv^2}{r}$$

As speed increases, more of gravity is “consumed” by the centripetal role, leaving less for N . At $v = \sqrt{gr}$, $N = 0$ — gravity alone fully accounts for the centripetal demand, with nothing left over. The car is momentarily weightless.

2.5 Applying the Model: The Valley

At the bottom of a valley, the center of the circular path is *above* the car. But gravity only ever points *down* — it physically cannot point toward a center that is above the object. Gravity *cannot* serve as the gravity-string here.

If gravity cannot supply the centripetal demand, yet a centripetal force is still required (the car is still moving along a curve), the model demands that the *sign* of the gravity-string term flip:

$$mg = N - \underbrace{\frac{mv^2}{r}}_{\text{gravity-string, now subtracted}} \quad \Rightarrow \quad N = mg + \frac{mv^2}{r}$$

The normal force must now *exceed* mg — supplying both the weight sensation *and* the centripetal demand, since gravity offers no help in this geometry. This is the “pressed into your seat” sensation at the bottom of a roller coaster dip.

2.6 The Resolution

The unifying statement:

mg is a fixed total. The normal force N — the sensation of weight — is whatever remains of that total after the centripetal demand $\frac{mv^2}{r}$ has been accounted for. Whether the centripetal demand *subtracts from* or *adds to* N depends entirely on whether gravity’s fixed downward direction *aligns with* or *opposes* the direction toward the center of curvature at that point in the path.

This single framework — one fixed quantity, mg , redistributed between “felt weight” and “centripetal demand,” with the redistribution’s sign set by geometry — replaces two memorized formulas with one mental picture from which both formulas, and their correct signs, can be derived on demand.

2.7 Historical and Disciplinary Context

The physical content of the gravity-string model — that the sensation of weight is precisely the normal force, and that this sensation can be reduced, eliminated, or amplified by acceleration, without gravity itself changing at all — is a special case of what Einstein formalized in 1907 as the **Equivalence Principle**, the foundational idea behind General Relativity: that gravity and acceleration are, locally, indistinguishable. An observer in a sealed box cannot determine, by any local experiment, whether they are at rest in a gravitational field or accelerating in the absence of one.

The vertical circular motion equations themselves — $N = mg \mp \frac{mv^2}{r}$ — are standard results in introductory mechanics, typically derived by directly applying Newton’s second law in the radial direction with no unifying narrative connecting the two cases. The “Gravity-String” framing — a single conserved quantity mg , split between two roles, with the split’s sign determined by the geometric relationship between gravity’s fixed direction and the variable direction toward the center of curvature — was developed independently as a device for remembering and re-deriving both signs from one picture, without prior exposure to the Equivalence Principle or its history.

3 The Virtual Two-Step Transfer Model

3.1 The Question That Started It

In an elastic collision between two objects of equal mass, the result is famous and strange: the two objects *swap velocities* entirely. The algebra confirms this — solving the two conservation equations simultaneously gives $v_{1f} = v_{2i}$ and $v_{2f} = v_{1i}$ — but the algebra alone does not explain *how* or *why* a clean swap occurs. What is physically happening during the collision that produces this exact result?

This model has two parts: a **virtual two-step transfer**, describing how an elastic collision can be decomposed into two imagined sequential momentum transfers; and a **drift/blob model**, describing what physically happens to the “missing” kinetic energy in an inelastic collision.

3.2 Part One: The Virtual Two-Step Transfer (Elastic Collisions)

Consider two objects of equal mass, A moving at $+v$ and B moving at $-v$ (approaching each other symmetrically). The collision is instantaneous in reality — but its *net effect* can be reproduced by imagining two sequential virtual steps:

Step 1 — A transfers its momentum to B .

Virtually, A 's momentum ($+v$) is handed entirely to B . A comes to rest (virtual state: $v_A = 0$). B receives the transfer (virtual state: $v_B = -v + v = 0$, momentarily at rest as well, having received a momentum that exactly cancels its own).

Step 2 — B transfers its (original) momentum to A .

B 's *original* momentum ($-v$) is virtually handed to A . B comes to rest. A receives $-v$.

Combining the two virtual steps:

A ends at $-v$ (received in step 2). B ends at $+v$ (received in step 1, before its own original momentum was virtually removed in step 2 — net effect: B ends carrying A 's original momentum).

The result is a complete velocity swap — exactly matching the algebraic solution, $v_{Af} = v_{Bi}$ and $v_{Bf} = v_{Ai}$.

The two steps are explicitly *virtual* — they do not occur in sequence in real time; the actual collision is instantaneous. But the two-step decomposition produces the correct final state, and gives a transfer-based narrative for *why* a swap occurs: each object's momentum is, in effect, fully exchanged with the other's.

When the two equations governing an elastic collision — conservation of momentum, and the relative-velocity (“approach equals separation”) relation that replaces the more cumbersome kinetic energy equation — are solved with these two unequal masses, the velocities are not a simple swap, but the same two-step transfer logic still describes the *direction* of each exchange; only the *magnitudes* of what is transferred in each virtual step now depend on the mass ratio.

3.3 Part Two: The Drift / Blob Model (Inelastic Collisions and Explosions)

The second half of the model addresses a different question: in a collision (or its time-reverse, an explosion) where kinetic energy is *not* conserved, where does the missing energy go, and how can momentum remain exactly conserved while kinetic energy changes?

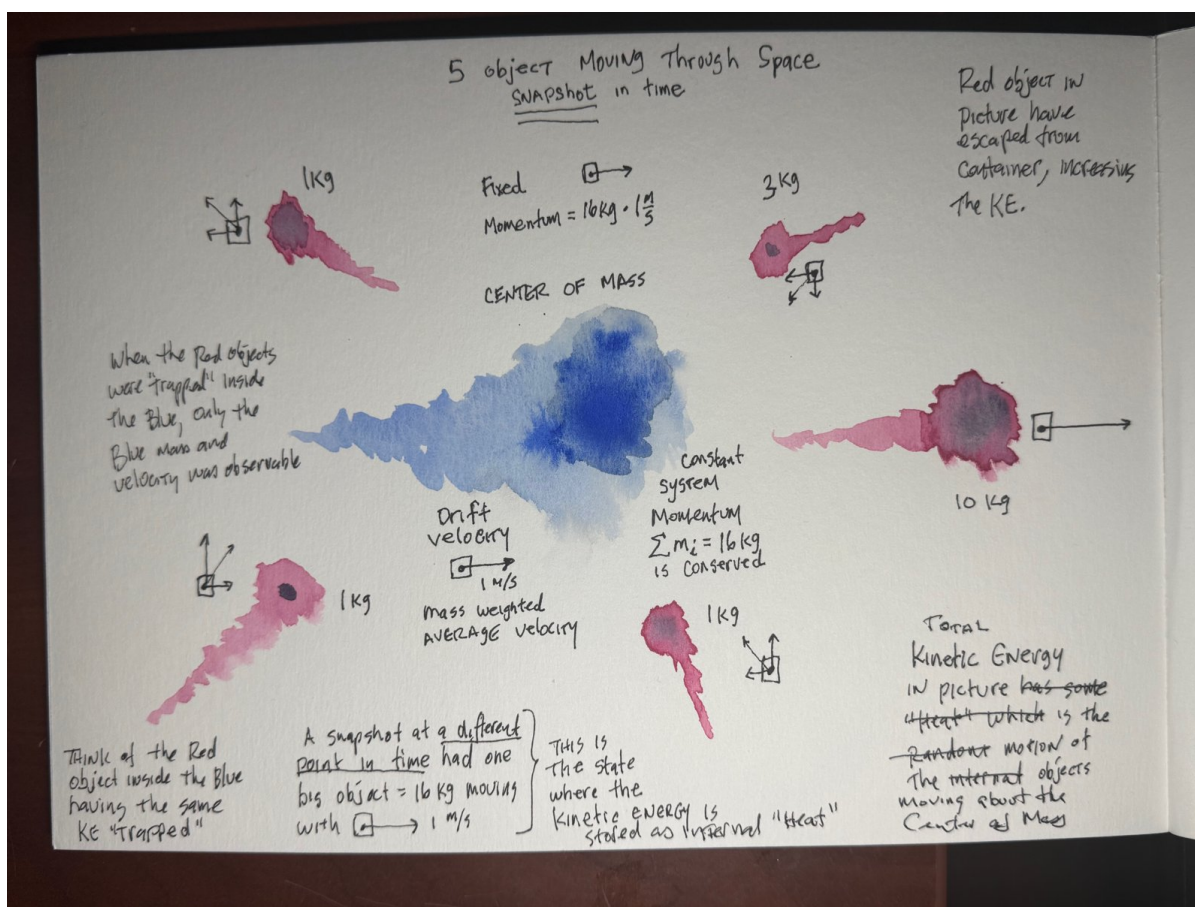
The picture: imagine the colliding (or exploding) objects as a single “blob” — one object with a well-defined center of mass (COM) and a single COM velocity, v_{cm} , determined entirely by total momentum and total mass:

$$v_{cm} = \frac{p_{total}}{m_{total}}$$

The explosion case (drift + internal motion):

A single object at rest (or moving with some v_{cm}) explodes into fragments. **The COM of the fragments continues moving at exactly v_{cm} , unchanged** — this is simply conservation of momentum, restated. But the *individual* fragments now also have velocity *relative to the COM* — they are “speeding around” the drifting center.

The *total* momentum is unchanged (still $m_{total} \cdot v_{cm}$, since momenta relative to the COM sum to zero by definition of the COM). But the *total kinetic energy* has *increased* — by exactly the kinetic energy of the fragments’ motion *relative to the COM*. That relative-motion energy came from the explosion’s energy source (e.g., a chemical reaction) being converted into kinetic energy.



An original sketch of the explosion case: five fragments (masses labeled, 1–10 kg, summing to $m_{total} = 16 \text{ kg}$) escape from the central blue “center of mass” blob, each carrying its own velocity vector. The blob’s drift velocity

($v_{cm} = 1 \text{ m/s}$) is the mass-weighted average of all five — conserved exactly, regardless of how the fragments scatter. The kinetic energy of each fragment’s motion relative to that drift is conceptualized as “trapped” or internal kinetic energy — the same quantity that, run in reverse, becomes heat in a collision.

The inelastic collision case (the reverse):

Multiple objects, each with their own velocity, collide and stick together into one “blob.” Before the collision, each object has some velocity *relative to the eventual COM*. After the collision, by definition, the blob is rigid — there is no more relative motion between the pieces. **All of the kinetic energy that existed as motion relative to the COM before the collision is gone after the collision** — it has been converted to heat and deformation, conceptualized as random internal motion “inside the blob.”

Meanwhile, the COM velocity itself — $v_{cm} = p_{\text{total}}/m_{\text{total}}$ — is exactly the same before and after, because it depends only on the conserved total momentum and the (also conserved) total mass. **Momentum is conserved because the COM motion never changes. Kinetic energy decreases because the relative motion is destroyed.**

3.4 The Elegant Capstone

The drift/blob model leads directly to the same conclusion reached independently via the momentum–energy parabola in Chapter 1, but by an entirely different route:

When is *all* kinetic energy lost in a perfectly inelastic collision?

Only when the relative motion accounts for *all* of the kinetic energy — that is, only when $v_{cm} = 0$, so that the COM itself carries *zero* kinetic energy to begin with. And $v_{cm} = 0$ requires $p_{\text{total}} = 0$.

If the COM is moving at all, the blob retains that motion after collision — some kinetic energy survives, by definition, as the kinetic energy of the COM’s own motion. Total loss of kinetic energy is possible *if and only if* the COM was at rest — i.e., the total momentum was zero — before the collision.

This is the identical conclusion as the parabola’s minimum touching zero only when $p_{\text{total}} = 0$ — reached here through a physical narrative about drift and internal motion, rather than through the geometry of a constrained quadratic. The two models — one geometric, one physical/narrative — are different lenses on the same underlying fact, and their agreement is itself a form of validation: two independently constructed mental models, built at different times and from different starting points, converge on the same boundary condition.

3.5 Historical and Disciplinary Context

Part One (virtual two-step transfer): The end result — equal-mass elastic collisions produce a complete velocity exchange — is a standard, well-known result, often demonstrated with Newton’s cradle. However, the *two-step sequential transfer* framing, presented as two imagined (virtual, non-sequential-in-reality) hand-offs of momentum between the two bodies, does not appear in standard introductory or intermediate mechanics texts surveyed. The standard treatment simply solves the two simultaneous equations and observes the swap as an algebraic consequence. The two-step transfer narrative appears to be an independent pedagogical invention — a physical story for an algebraic result that is usually presented without narrative at all.

Part Two (drift/blob model): The underlying physics — that kinetic energy decomposes exactly into “COM kinetic energy” plus “kinetic energy relative to the COM,” and that inelastic collisions destroy only the latter — is a known and important result in mechanics, sometimes called the **König’s theorem** decomposition of kinetic energy (after Johann Samuel König, 18th century), and is foundational to the treatment of rigid body dynamics and multi-particle systems. The decomposition is standard; the “blob with internal drift” narrative framing, and its use as an intuitive explanation accessible without the formal König’s theorem machinery, was developed independently.

The convergence of this model’s conclusion with the momentum–energy parabola’s conclusion (Chapter 1) — both independently arriving at “ $p_{\text{total}} = 0$ is necessary and sufficient for total KE loss in a perfectly inelastic collision” — mirrors the fact that König’s theorem and the COM-frame parabola analysis are, in the formal literature, two views of the same underlying mathematics.

4 Momentum as State: The Center-of-Mass Split and Noether's Theorem

4.1 The Question That Started It

After building the drift/blob model in Chapter 3, one question remained unresolved: *is momentum a form of energy?* It is conserved, the way energy is conserved. It participates in collisions, the way energy does. And yet kinetic energy can convert to heat while momentum cannot be “lost” to heat in the same way. If momentum is not a form of energy, what *is* it — and how does it relate to energy?

This question led to a much deeper one, reached by extending the drift/blob model with a property called “heat” representing all internal disordered motion, and asking: *why* is momentum conserved at all? Is it an empirical fact about the universe, discovered by observation — or is it, in some sense, *built into the definition* of an isolated system?

4.2 Refining the Blob: Center-of-Mass KE vs. Rest-Frame KE

The drift/blob model from Chapter 3 can be made precise with one additional idea: **every system of particles has a center of mass (COM)**, and its velocity is fixed entirely by the total momentum:

$$v_{cm} = \frac{p_{total}}{m_{total}}$$

The total kinetic energy of any system of particles splits *exactly* into two pieces — not approximately, but as an algebraic identity:

$$KE_{total} = \underbrace{\frac{1}{2}m_{total}v_{cm}^2}_{\text{COM KE}} + \underbrace{KE_{rel}}_{\text{“rest-frame” KE}}$$

The KE of the whole system drifting together, plus the KE of motion *around* that drift.

COM KE is the kinetic energy the system would have if every particle moved at exactly v_{cm} — the energy of the “whole blob drifting together.”

Rest-frame KE (also called relative KE) is the kinetic energy measured by an observer riding along *with* the center of mass, who sees the COM as stationary and watches only the particles’ motion *relative to it*.

The train analogy:

Picture a moving train. From the ground, a passenger walking down the aisle has two components to their motion: the train’s speed (the “drift”) and their own walking speed relative to the train car (the “rest-frame” motion). An observer on the ground sees both combined. An observer seated on the train sees only the walking. **COM KE is the “train” part. Rest-frame KE is the “walking” part.**

4.3 Why the Split Matters: One Part Is Locked, the Other Is Free

Because $v_{cm} = p_{total}/m_{total}$, and p_{total} is conserved in an isolated system (Chapters 1 and 3), **the COM KE is fixed** — it cannot change, no matter how violent or sticky the collision is. It is determined entirely by the conserved quantity p_{total} .

The rest-frame KE is the only part that can change. This is precisely the quantity that converts to heat, sound, and deformation in an inelastic collision.

The corrected statement of the Chapter 3 result:

A perfectly inelastic collision converts *all* of the rest-frame KE into heat — always, by definition of “perfectly inelastic” (no relative motion remains). **The final blob is at rest ($KE_{total,final} = 0$) only if the COM KE was already zero to begin with** — i.e., only if $v_{cm} = 0$, i.e., only if $p_{total} = 0$. If $p_{total} \neq 0$, the COM KE survives the collision intact, no matter how inelastic the collision is, and the final blob drifts onward carrying exactly that much kinetic energy.

This is the same $p_{total} = 0$ conclusion reached in Chapter 1 (the parabola minimum) and Chapter 3 (the drift/blob model) — now derived a third, independent way, from the algebraic identity $KE_{total} = KE_{cm} + KE_{rel}$. Three different routes to the same destination is strong evidence the destination is correct.

4.4 Is Momentum an Energy, or Something Else?

The resolution to the original question: **momentum is not a form of energy. It is a state variable** — in the same category as position or velocity, not in the same category as kinetic energy, potential energy, or heat.

Momentum	Energy (KE, heat, etc.)
Vector (has direction)	Scalar (no direction)
Units: kg·m/s	Units: Joules
Describes <i>how</i> a system is moving as a whole	Describes <i>how much</i> motion/energy exists
Fixes v_{cm} , hence fixes KE_{cm}	Can move between forms (KE $\leftrightarrow$ heat)

The clean statement:

Momentum constrains *which* states are reachable. Energy describes *how much* can move between forms within that constraint. Momentum doesn’t “become” energy in a collision — but by fixing v_{cm} , it fixes exactly how much of the total KE is locked into COM motion and therefore *cannot* become heat, regardless of how the collision proceeds.

4.5 The Deeper Question: Why Is Momentum Conserved At All?

Pushing one level deeper: within a single fixed- p_{total} table (Chapter 1’s spreadsheet), momentum conservation is trivially true *by construction* — every row was built to satisfy it. But this raises an

uncomfortable question: *if momentum conservation is just how the table was built, in what sense is it a law of nature at all, rather than a definition we imposed?*

The resolution: **it is both** — and the relationship between the two is one of the most important ideas in all of physics.

Step 1. Define an “isolated system” as one with no external forces acting on it.

Step 2. Observe that the laws of physics do not depend on *where* in space an experiment is performed — a collision experiment gives the same result in a lab in California as in a lab on the Moon, all else equal. This is called **translational symmetry of space**.

Step 3 (Noether’s theorem). Every continuous symmetry of this kind corresponds to a conserved quantity. Translational symmetry of space corresponds to **conservation of linear momentum**.

The synthesis:

Conservation of momentum *is* a consequence of how an isolated system is defined — *combined with* a deep symmetry of space itself. It is not “just” a definition (it follows necessarily from the symmetry, via Noether’s theorem), and it is not “just” an empirical observation (the empirical observation *confirms* that this symmetry actually holds in our universe). It is a definition that becomes a law *because* of a symmetry, and that law is then verified, to extraordinary precision, by experiment.

This three-step structure — (1) define isolation, (2) identify a symmetry, (3) derive a conservation law via Noether’s theorem — is the same structure underlying *every* conservation law in physics:

Symmetry of the laws of physics	Conserved Quantity
The laws don’t change over time	Energy
The laws are the same everywhere in space	Linear Momentum
The laws are the same in every direction	Angular Momentum

4.6 Historical and Disciplinary Context

The center-of-mass / rest-frame KE decomposition is the same König’s theorem result referenced in Chapter 3, here derived through a different route: by extending the drift/blob model with the explicit algebraic identity $KE_{\text{total}} = KE_{\text{cm}} + KE_{\text{rel}}$, and by introducing “heat” as a placeholder for all forms of internal/rest-frame energy. This identity is standard in any treatment of multi-particle systems and is the starting point for the entire field of statistical mechanics, where “rest-frame KE” becomes, in aggregate, the formal definition of **temperature**.

Noether’s theorem (Emmy Noether, 1918) is one of the most important results in all of theoretical physics. It is the rigorous foundation for the statement “momentum conservation follows from translational symmetry of space,” and the analogous statements for energy (time symmetry)

and angular momentum (rotational symmetry). Noether's theorem underlies classical mechanics, quantum mechanics, and quantum field theory alike.

The classical mechanics tradition that builds physics from symmetry first — identifying symmetries of a system and deriving conserved quantities from them, rather than treating conservation laws as separate empirical facts to be memorized — is most closely associated with the Lagrangian and Hamiltonian formulations of mechanics (18th–19th century, Lagrange and Hamilton), and is given an especially clear, symmetry-first exposition in *Mechanics* by Landau and Lifshitz (1960s), the first volume of their famous course in theoretical physics.

What was independently constructed here: the spreadsheet's structure — fixed- p tables as a discrete, concrete “state space,” with momentum imposed *as the defining constraint* of each table and KE explored *as the variable within that constraint* — is a small-scale, hands-on instance of exactly the state-space-plus-symmetry logic that Noether's theorem formalizes and that Landau and Lifshitz use as the organizing principle for all of classical mechanics. The path from “I built a spreadsheet and noticed momentum was constant by construction” to “this is what Noether's theorem says” was walked without prior exposure to either Noether's theorem or the Landau–Lifshitz tradition — arriving, independently, at the same place those two long-standing traditions in theoretical physics occupy.

5 Reverse-Engineering the Painting: A New Way to Practice

5.1 The Question That Started It

The watercolor sketch in Chapter 3 (the explosion case of the drift/blob model) was drawn to illustrate an idea: a center-of-mass “blob” drifting at constant velocity, with fragments escaping at various speeds and angles, their momenta summing back to the blob’s total. But the sketch as drawn left the fragment velocities *unlabeled* — only the masses (10, 3, 1, 1, 1 kg) and the blob’s drift velocity (1 m/s, purely horizontal) were given, along with rough directional arrows.

This was not an oversight. It was the beginning of a different question: *could the sketch itself — with its masses and directions already drawn, but velocities left blank — be used as a worksheet?* Could a student look at the picture, know only the total momentum it must add up to, and *reverse-engineer* a complete, self-consistent set of velocities using nothing but guess-and-check and a running total?

5.2 A Preliminary: Mass Is Fixed (For Our Purposes)

Before working through the figure, one simplification is worth stating explicitly. The figure labels the blue COM blob with a mass of 16 kg — exactly the sum of the red fragments’ masses ($10 + 3 + 1 + 1 + 1 = 16$). This is intentional, and it reflects a deliberately **Newtonian** simplification: mass is treated as fixed and additive, independent of velocity.

Special relativity tells us that an object’s measured mass increases with its velocity relative to an observer. At everyday speeds — even very fast everyday speeds, like a car or a bullet — this effect is so small it is completely undetectable, and Newtonian mechanics (treating mass as fixed) is exactly correct for all practical purposes. The relativistic correction only becomes significant at speeds approaching the speed of light, which is the domain of particle accelerators and astrophysics, not collision problems on a problem set. For this entire document, as in essentially all of AP Physics 1 and introductory mechanics, mass is simply a fixed number attached to each object.

This is what allows the figure’s “container” framing: while the fragments are inside the blue blob, the blob simply *has* 16 kg of mass. Once the fragments escape, the blue blob becomes a *virtual* construct — a mathematical center of mass with no physical substance of its own — but the 16 kg total is unchanged, because mass was never lost or gained; it was only ever the sum of the fragments’ masses, before and after.

5.3 The Setup: What the Figure Gives You, and What It Withholds

Looking at the figure: the blue blob has an explicit velocity, $v_{cm} = 1$ m/s, purely in the $+x$ direction (no y -component is drawn for the blob). Each red fragment has a labeled mass and a *direction* (drawn as an arrow), but **no velocity is labeled on any fragment**. This is the deliberate gap the figure leaves for the reader to fill in.

The target, stated as two numbers:

Because $v_{cm} = 1$ m/s in the $+x$ direction only, and $m_{total} = 16$ kg, the *total* momentum of the

system is fixed:

$$p_{x,\text{total}} = m_{\text{total}} v_{cm,x} = 16 \times 1 = 16 \text{ kg}\cdot\text{m/s} \quad p_{y,\text{total}} = m_{\text{total}} v_{cm,y} = 16 \times 0 = 0 \text{ kg}\cdot\text{m/s}$$

Every fragment's individual momentum — mass times velocity, in each direction — must sum to exactly these two numbers. That is the *only* constraint. Everything else is free.

5.4 Step 1 — Bootstrap from the Largest Fragment (x-direction)

Working the x -direction first: start with the largest fragment, the 10 kg object, whose arrow in the figure points purely in $+x$ with no y -component (its own small free-body diagram in the sketch shows this directly). **Pick a velocity for it** — any value. Choose $v_x = +10 \text{ m/s}$.

Its momentum contribution: $p_x = mv_x = 10 \times 10 = +100 \text{ kg}\cdot\text{m/s}$.

The running total, after one fragment:

Target: $p_{x,\text{total}} = +16$. Current running total: $+100$. **The remaining four fragments must together contribute** $16 - 100 = -84 \text{ kg}\cdot\text{m/s}$ of x -momentum — a large *negative* correction. Since several fragments in the figure point in the $-x$ direction, this is achievable — and the size of the correction needed is itself informative: it tells you immediately that the first guess (10 m/s) was large enough to require real work from the other fragments, which builds intuition for how momentum scales with both mass and velocity.

5.5 Step 2 — The Next Fragment, and the Role of Components

The next-largest fragment is the 3 kg object. Unlike the 10 kg fragment, its arrow in the figure is *not* purely horizontal — the sketch shows it with both an x -component and a y -component (the small diagram with three arrows: one along the direction of motion, and two perpendicular arrows showing the split into components). This is the figure's visual introduction to **vector components** — without requiring any formal linear algebra to recognize what's being shown: *a single arrow can be thought of as “part sideways” and “part up-or-down,” and those two parts can be worked out separately.*

Since the -84 remaining x -momentum must be shared among four fragments (3 kg, 1 kg, 1 kg, 1 kg), and the 3 kg fragment points in $-x$, a reasonable guess is to assign it a *portion* of that -84 — not all of it, since three more fragments still need their turn. Suppose the 3 kg fragment is assigned $p_x = -54 \text{ kg}\cdot\text{m/s}$, leaving $-84 - (-54) = -30$ for the remaining three 1 kg fragments.

$$v_x \text{ for the 3 kg fragment} = \frac{p_x}{m} = \frac{-54}{3} = -18 \text{ m/s}$$

5.6 Step 3 — Completing the x-Components

Continuing this process — at each step, looking at the figure's arrows to decide which fragments should contribute positively or negatively, and choosing a value that keeps the running total on track toward $+16$ — one possible complete assignment is:

Fragment	v_x (m/s)	$p_x = mv_x$
10 kg	+10	+100
3 kg	−18	−54
1 kg	−20	−20
1 kg	−20	−20
1 kg	+10	+10
Total	—	+16 ✓

The total matches the target exactly. The x -direction is solved.

5.7 Step 4 — The y -Components, Starting Fresh

Because the blob’s velocity has *no* y -component, the target for the y -direction is simply $p_{y,\text{total}} = 0$. The same bootstrap process applies, but now starting from a different fragment — the figure shows the 3 kg fragment’s arrow with a clear y -component (this is the same arrow whose x -component was used above; the same arrow has both parts, and they are worked out independently).

A first guess of $v_y = -100$ m/s for the 3 kg fragment is informative *precisely because it is too large*: $p_y = 3 \times (-100) = -300$, which would require the remaining three 1 kg fragments to together contribute +300 — an enormous amount for objects with only 1 kg of mass each. **This “failed” guess is not wasted effort** — it demonstrates, concretely, that momentum depends on *both* mass and velocity, and that a small mass needs a correspondingly large velocity to make a meaningful contribution. A smaller guess, such as $v_y = -10$ for the 3 kg fragment ($p_y = -30$), is far more workable.

Fragment	v_y (m/s)	$p_y = mv_y$
10 kg	0	0
3 kg	−6.67	−20
1 kg	−10	−10
1 kg	−10	−10
1 kg	+40	+40
Total	—	0 ✓

The lesson hiding in the last row:

The final 1 kg fragment requires $v_y = +40$ m/s — by far the largest speed in either table, attached to one of the smallest masses. This is not a contrived result; it is what the arithmetic *demand*s once the other four values are chosen. **Seeing this number emerge naturally — as the thing needed to make the books balance — builds an intuitive feel for momentum as “mass-weighted velocity” far more effectively than being told the definition and given a formula.** A small mass genuinely needs a large velocity to matter; the arithmetic makes this unavoidable, rather than asserted.

5.8 What Was Just Demonstrated

The complete picture:

Starting from *only* the figure’s masses, directions, and the COM’s velocity, a complete, self-consistent, physically valid set of five velocity vectors was constructed — using nothing beyond

arithmetic, a running total, and the figure's drawn arrows as a guide for sign and rough proportion. **Both the x and y totals check out exactly against the required p_{total} .** This is a fully worked two-dimensional momentum problem, solved *backward* — from the answer to the inputs — using guess-and-check.

5.9 The State-Space Connection

The values chosen above are not the *only* valid answer. Any other set of five velocities — in both x and y — that produces the same two totals ($p_x = +16$, $p_y = 0$) would be *equally valid*. The figure's masses and the COM's velocity fix the total momentum; they do **not** fix the individual fragment velocities. There are infinitely many self-consistent assignments.

This is the *same* state-space idea encountered twice already in this document — the momentum–energy parabola of Chapter 1 (every point on the parabola is a valid state consistent with a fixed p_{total}) and the fixed- p spreadsheet tables of Chapter 4 (every row is a valid state). Here, the “state space” is the space of all possible five-fragment velocity assignments consistent with one drifting COM — and the figure, plus a target number, is enough to explore that space by hand.

A possible reordering of how this is taught:

The formal vector/component machinery — decomposing a vector into x and y parts, summing components separately, recombining with the Pythagorean theorem — is usually introduced *before* students are asked to reason about systems of multiple objects and conservation laws. The exercise above suggests an alternative order is possible: **the need for components arises naturally, and is motivated concretely**, once a student is trying to make several objects' momenta add up to a required total in two directions independently. The abstract tool (components) emerges as the answer to a concrete question (“how do I handle an arrow that points diagonally, when I need to track x and y totals separately?”), rather than being presented as a prerequisite skill to be mastered in isolation first. Whether this reordering would work well for a full class of students is an open question — but the *exercise itself*, working backward from a figure to a complete set of consistent velocities, does not require components to be taught first; it can *motivate* them.

5.10 A New Kind of Practice

The claim, stated carefully:

A worksheet of ten momentum problems typically asks a student to apply the *same* procedure ten times, with different numbers each time. The exercise above is different in kind: **the figure itself — one picture, with masses and directions but blank velocities — can generate an unlimited number of valid problems**, because any self-consistent assignment of velocities is a correct “answer,” and *checking* self-consistency (do the totals match the required p_{total} ?) is itself the entire skill being practiced.

This is not a claim that practice is unnecessary — repetition that builds genuine pattern recognition is valuable. The claim is narrower: **repetition that consists of running the same procedure on superficially different numbers, where the student already knows the procedure and is merely executing it, builds comparatively little new understanding.** The reverse-engineering exercise above replaces that kind of repetition with something closer to *design* — choosing values, checking consequences, revising — which is a different and arguably

more durable cognitive activity, and one typically reserved for advanced or research-level work rather than introductory practice.

5.11 Looking Forward

This figure-as-worksheet technique — a single illustration, with some quantities given and others left for the student to determine through guided guess-and-check against a known total — is, as far as can be determined, **not a standard feature of introductory physics instruction**. Textbooks provide formulas and worked examples; they do not, typically, provide a single rich illustration designed to be *filled in* by the student through iterative reasoning, with the figure itself serving as both the problem statement and the scratch pad.

A note on context. This technique, the four models in earlier chapters, and the website (physicsexplorer.com) referenced throughout this document were developed during an intensive period of self-directed learning, originally undertaken while developing a cutting edge, teaching platform for learning AP Physics 1. The material developed during that process — this document, the accompanying spreadsheet, and the interactive simulations on the website — is offered here as a record of what that process produced, and as a foundation that could, with further development, support a complete learning system: interactive simulations paired with reverse-engineering exercises of the kind demonstrated in this chapter, designed explicitly around *transparent* use of AI as a learning partner — a tool for building understanding through dialogue and iteration, rather than a shortcut to answers. The path this document has tried to model throughout — sketch, guess, check, revise, and only afterward connect to the formal literature — is itself the curriculum.

Conclusion

What These Four Models Have in Common

Each of the four models in this document arose from the same underlying move: taking a formula that was already correct and already known, and asking “*what physical picture would make this formula feel inevitable, rather than arbitrary?*”

- The **momentum–energy parabola** turned three separately-memorized collision formulas into a single curve with a clipped region, a minimum, and two marked points.
- The **gravity-string model** turned two formulas with opposite signs into one fixed quantity, split two ways, with the split’s direction set by geometry.
- The **virtual two-step transfer and drift/blob models** turned an algebraic swap and an abstract energy-loss formula into narratives about handing off momentum and about a drifting, internally-jittering blob.
- The **center-of-mass split and Noether’s theorem connection** turned the question “is momentum a form of energy?” into a clean state-versus-energy distinction, and turned “why is momentum conserved?” into a three-step chain from symmetry to conservation law — the same chain that underlies every conservation law in physics.

A fifth contribution, Chapter 5, is different in kind from the four above: rather than a mental model for understanding *one* formula, it is a **technique for generating practice itself** — turning a single illustration into an open-ended, self-checking exercise, and turning the state-space idea (already encountered twice, in Chapters 1 and 4) into something a student can manipulate directly with pencil and arithmetic.

Why This Matters for Learning

“When we understand a problem it always appears simple — that does not mean it was not complex. It means we understand it.”

A formula that is merely memorized is fragile: it can be misapplied, mis-signed, or forgotten under exam pressure. A formula that is attached to a *picture* — a parabola with a clipped arc, a fixed budget split two ways, a blob with a drifting center and a jittering interior — can be *re-derived* on the spot, because the picture itself constrains what the answer must look like.

This is not a claim that intuition can replace rigor, or that these models are substitutes for the formal derivations found in upper-division texts. It is, instead, a demonstration that the *order* can be reversed: a careful learner, working from algebra, arithmetic, sketches, and an insistence on physical sense-making, can arrive at — or very near — the same conceptual territory that formal derivations occupy, *before* ever encountering the formal version.

For Students

If you are reading this as a student: every one of these models began with the same sentence — “*I can do the math, but I don’t understand why it works.*” That sentence is not a sign of confusion to be embarrassed about. It is the correct starting point for genuine understanding. Sketch it. Build a small table of numbers and look for a pattern. Invent a story — even a slightly silly one, like a virtual string

or a drifting blob — and then check whether the story predicts the same numbers the formula does. If it does, you have built something durable. If it doesn't, the mismatch will show you exactly what you misunderstood.

This is how all four models in this document were built. There is no reason the same path is not available to you.

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